EMC Lab 1: Crosstalk

Laboratory Report

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1 Introduction

In any electromagnetic interference problem there are three basic elements involved: the source of disturbance the system in which the interference is observed and the medium of coupling between the two elements. When the source of disturbance and the affected system are in close proximity, the electromagnetic interference is referred to as *crosstalk*. In this experiment the crosstalk between two parallel wires separated by a common ground plane and placed in a Perspex dielectric with $\varepsilon_r = 2.25$ is measured using pulse excitation for frequencies where the wires behave as both electrically short and electrically long.

Two distinct coupling mechanisms are considered separately and in combination:

- **Capacitive coupling:** occurs between two wires through the electric field and is associated with voltage changes $\left(\frac{dV}{dt}\right)$ on the driven wire.
- **Inductive coupling:** occurs between two wires through the magnetic field and is associated with current changes $\left(\frac{dI}{dt}\right)$ on the driven wire.

The termination condition of the driven wire (Line 1) is the controlling factor in the relative importance of the two coupling mechanisms, as summarised in Table 1:

Termination	**Dominant Mechanism**	**Reason**
Open circuit	Capacitive only	Isolates electric field coupling
Short circuit	Inductive only	Isolates magnetic field coupling
Matched load	Combined	Both mechanisms present

Table 1: Effect of termination condition on coupling mechanism.

In this experiment, a pulse generator (PM5786A) and a Keysight DSOX1202G oscilloscope are used to measure the coupled voltages on the adjacent wire.

2 Results and Analysis

2.1 Line Electrical Length:

The wave propagation speed in the Perspex dielectric is:

$$v = \frac{c}{\sqrt{\varepsilon_r}} = \frac{3 \times 10^8}{\sqrt{2.25}} = 2 \times 10^8 \text{ m/s} \tag{1}$$

The wavelength and electrical length of the $\ell = 1.064$ m line are calculated as $\lambda = v/f$ and ℓ/λ respectively, and summarised in Table 2:

Frequency	λ (m)	ℓ/λ (%)	Electrically Short?
1 MHz	200	0.53	Yes $(\ell \ll \lambda/10)$
100 MHz	2	53	No $(\ell > \lambda/10)$

Table 2: Wavelength and electrical length of the line at two frequencies.

At 1 MHz the lumped circuit approximation is valid and the simple capacitive and inductive coupling models can be used. At 100 MHz the line has to be treated as a distributed system and the simple models cannot be used.

Preliminary Setup:

Why Not Use a T_Connector to Split the Output of Output A?

If a T_connector is used it will place the 50 Ω oscilloscope input in parallel with the 50 Ω transmission line cable resulting in a total equivalent load of:

$$R_{\text{eq}} = \frac{50 \times 50}{50 + 50} = 25\ \Omega \tag{2}$$

on the generator this is a severe impedance mismatch that causes a distorted waveform including a distorted rise time rendering all subsequent measurements invalid. By using Output B with the Invert function, this problem is avoided altogether.

What is Inside the Matching Box?

The matching box most likely contains a resistive L pad attenuator network that transforms the source impedance from 50 Ω to 115 Ω. This ensures impedance matching at the input of Line 1, such that the reflection coefficient is:

$$\Gamma = \frac{Z_L - Z_0}{Z_L + Z_0} = 0 \qquad \text{when} \quad Z_L = Z_0 = 115\ \Omega \tag{3}$$

resulting in no reflections at the input of the line.

2.2 Capacitive Coupling: Line 1 Open Circuit

With Line 1 open circuited at end D and the generator set to minimum transition times voltage spikes were observed on CH2 at each rising and falling edge of the square wave input. No crosstalk was observed during the flat portions of the waveform where $dV/dt = 0$. This directly verifies that capacitive crosstalk is proportional to dV/dt.

Amplitude Measurements

Quantity	Value
Generator output p2p (CH1)	7.8 V
Actual voltage on Line 1 (÷3)	2.6 V pp → 1.3 V peak
Coupled spike End D (CH2)	∼320 mV peak
Coupled spike End C (CH2)	∼250 mV peak

Table 3: Measured voltage amplitudes for capacitive coupling (Line 1 open circuit).

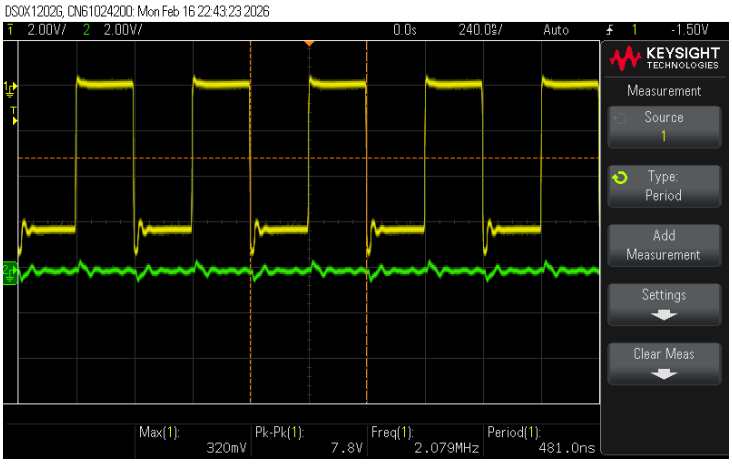


Figure 1: Capacitive coupling, line 2 End D (minimum rise time). CH1 (yellow): input square wave, CH2 (green): coupled spikes 320mV.

Percentage Coupling

$$\% \text{ Coupling} = \frac{V_{\text{cross}}}{V_{\text{Line1, peak}}} \times 100 = \frac{300}{1300} \times 100 \approx \boxed{23\%} \tag{4}$$

Transition Time Variation

The peak coupled spike was approximately 300 mV at the minimum transition time of $t_r = 5$ ns. Increasing the transition time to $t_r = 10$ ns halved the spike amplitude to ∼150 mV look Figure 2. This is consistent with the theoretical prediction:

$$V_{\text{cross}} \propto \frac{dV}{dt} = \frac{\Delta V}{t_{\text{rise}}} \tag{5}$$

The linear dependence of V_{cross} on dV/dt verifies the capacitive coupling theory of Equation 5.

Rise Time t_r (ns)	dV/dt (V/ns)	V_{cross} (mV)
5	0.52	300
8	0.33	190
10	0.26	150
15	0.17	100
20	0.13	75
30	0.08	50

Table 4: Measured coupled voltage as a function of rise time and dV/dt capacitive coupling.

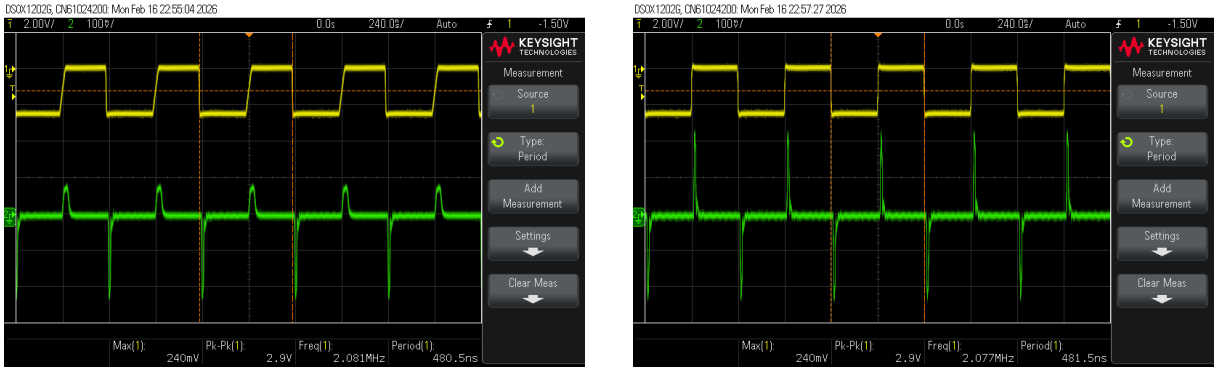


Figure 2: Capacitive coupling with increased rise time showing reduced spike amplitude

Coupling onto Lines 3 and 4

Line	Coupled Amplitude (mV)	% Coupling
Line 2 (adjacent)	300	23%
Line 3 (one gap away)	∼4	0.83%
Line 4 (two gaps away)	∼0	0%

Table 5: Capacitive crosstalk amplitude as a function of line separation.

The dramatic decrease in capacitive crosstalk with separation distance confirms that the mutual capacitance C_{12} falls off rapidly with distance as the electric field coupling between non adjacent lines is negligible.

2.3 Inductive Coupling: Line 1 Short Circuit

With Line 1 short circuited at End D the nature of coupling was different; the waveform on Line 2 showed larger pulses with different polarities at End C and End D indicating inductive coupling due to mutual inductance M, Figure 5.

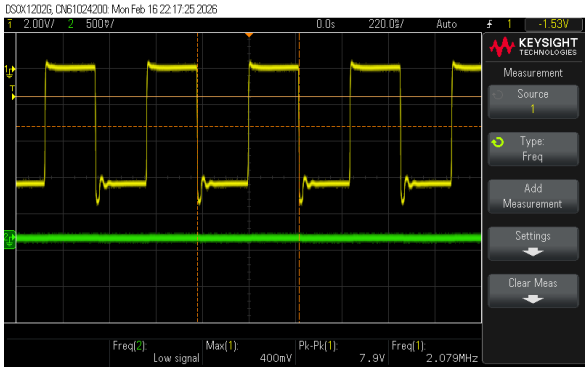


Figure 3: Capacitive coupling on line 3, showing negligible coupling (4mV)

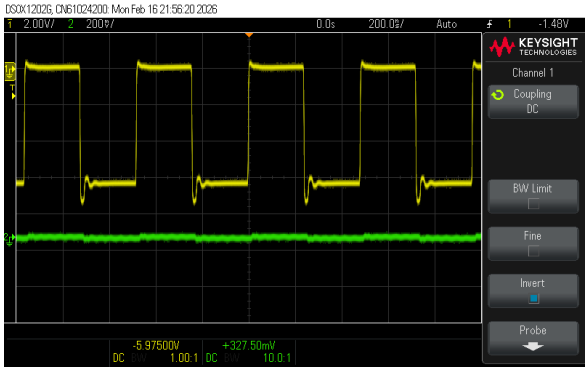


Figure 4: Capacitive coupling on line 4, essentially zero coupling

Amplitude Measurements

Quantity	Value
Generator output p2p (CH1)	2.9 V
Actual voltage on Line 1 ($\div 3$)	0.97 V pp $\rightarrow$ 0.48 V peak
Coupled spike End D (CH2)	$\sim$200 mV peak

Table 6: Measured voltage amplitudes for inductive coupling (Line 1 short circuit).

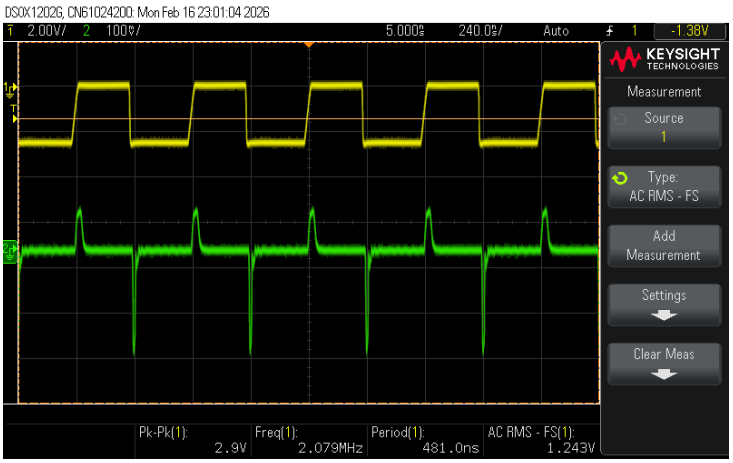


Figure 5: Inductive coupling, line 2 End D (line 1 shorted). Large bipolar pulses

Percentage Coupling

$$\% \text{ Coupling} = \frac{200}{480} \times 100 \approx \boxed{42\%} \tag{6}$$

The inductive coupling percentage was higher compared to the capacitive coupling percentage which was $\sim$23%. This was in agreement with the theoretical prediction; at minimum transition times the changes in the current are very rapid, resulting in a larger inductive coupling.

End C vs. End D

The amplitude of coupling at End C was larger compared to End D, indicating inductive coupling was larger at the point where the signal was launched on Line 1. Inductive coupling was larger at the near end crosstalk. This is because the rapidly changing current on line 1 produce its strongest magnetic field interaction at the launch end of the line ,as in Figure 6.

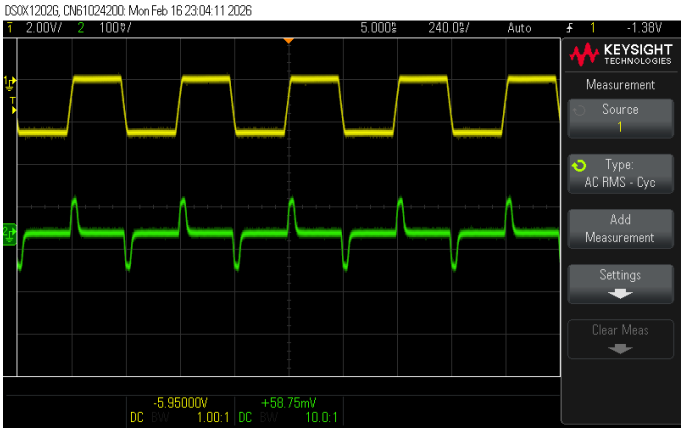


Figure 6: Inductive coupling, line 2 End C. Larger amplitude confirming NEXT

2.4 Matched Termination: (Line 1 = 115 Ω)

When a matched 115 Ω termination was applied to Line 1, the coupled waveform on Line 2 exhibited both capacitive and inductive coupling. A large difference in amplitude between End C and End D of Line 2 was observed:

End	Coupled Amplitude (V pp)
End D (far end)	∼460 mV
End C (near end)	∼40 mV

Table 7: Coupled amplitude at each end matched termination.

The amplitude at End D is approximately 11 times larger than that at End C. This is because capacitive and inductive coupling add at one end and subtract at the other. When a resistive termination is used on Line 1 both voltage and current are present on that line, the phase relationship between capacitive and inductive coupling determines whether they add or subtract at both ends ans showes in Figure 7 & 8.

3 Discussion

The results obtained using all three termination conditions match theory.

Capacitive coupling was verified by observing voltage spikes only at signal edges with no coupling occurring between flat regions. This is characteristic of $V_{\text{cross}} \propto dV/dt$. A measured coupling of **23%** on Line 2 dropped off sharply to **0.83%** on Line 3 and essentially zero on Line 4. This is consistent with the rapid decay of mutual capacitance C_{12}

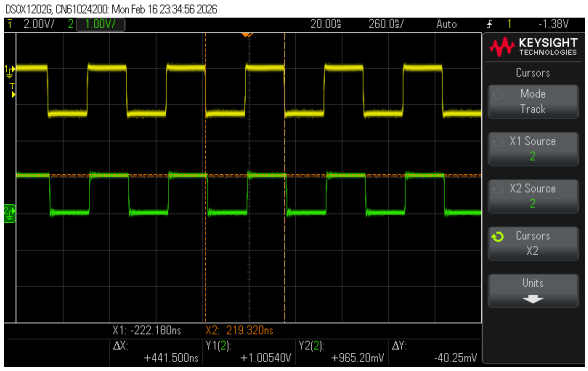


Figure 7: Matched termination, End D. 460mV coupled waveform

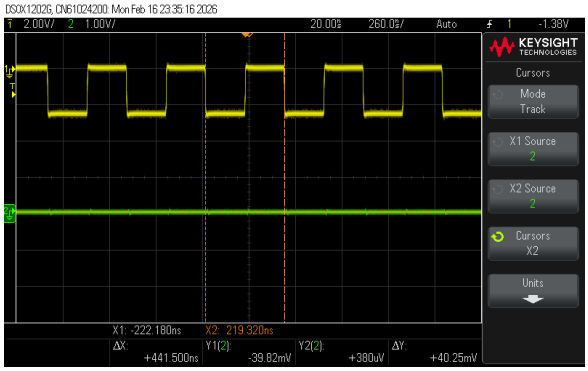


Figure 8: Matched termination, End C. 40mV showing partial cancellation

with wire separation. Doubling the rise time from approximately 5 ns to approximately 10 ns also halved the spike amplitude from 300 mV to 150 mV. This is exactly proportional to theory.

Inductive coupling also showed greater coupling (∼**42%**) than capacitive coupling (∼23%). This is again consistent with theory because $V_{\text{cross}} \propto -j\omega M/4Z_0$ and at $Z_0 = 115\ \Omega$ inductive coupling is greater. Also, End C had greater amplitude than End D. This is because of near end dominance, which is also predicted by theory because of greater magnetic field coupling to the current source.

The **matched termination** results were also asymmetrical. End D had 460 mV which is **11 times greater** than End C's 40 mV. The major sources of error were the voltage correction factor of approximately 1/3 via the matching box, ground lead inductance of the probe, and using an estimate rather than a measured value for the rise times in the dV/dt table.

4 Conclusion

This experiment demonstrated and quantified the two fundamental mechanisms of electromagnetic crosstalk between parallel transmission lines above a ground plane.

Capacitive coupling produced 23% coupling on Line 2 falling to 0.83% on Line 3 and essentially zero on Line 4, confirming the rapid decay with wire separation. Doubling the rise time from ∼5 ns to ∼10 ns halved the spike amplitude confirming $V_{\text{cross}} \propto 1/t_{\text{rise}}$.

Inductive coupling produced stronger coupling of ∼42% on Line 2 with larger amplitude at End C than End D confirming near end dominance.

With matched termination, End D (∼460 mV) was 11× larger than End C (∼40 mV), demonstrating constructive and destructive interference between the two coupling mechanisms at opposite ends of the other line.

All results were in good agreement with the circuit theoretical models confirming the validity of the electrically short approximation at the frequencies used. The experiment highlights why controlling both trace separation and signal edge rates is critical for managing crosstalk in high speed digital systems.